locate_tooth(FDI tooth)

returns a reference box (expert-verified box for annotated teeth; detector box otherwise)

Perturbation roll (fixed rates)

applied to the box SHOWN to the policy, never to the logged reference box

none - 45%

box unchanged
clean accept-and-proceed demonstration

small - 25%

offset 12-28% of box size, random sign
genuine judgment call: often still fine given zoom padding - never resolved by the prompt

big - 30%

offset 45-75% of box size, random sign
visually self-evident without any verbal cue

Teacher inspects the crop (zoom_crop)

and decides from the pixels alone whether the box is acceptable

nudge_crop

dx_frac, dy_frac, scale

zoom_crop on the corrected box - verify - commit finding

the same judgment the deployed policy faces at inference, with no ground truth available

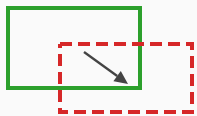
Decoupled from the detector

rates are configured constants, not the detector's real error rate, so demonstrations stay valid as the grounding tool improves.

No verbal cue

an error is never announced in text; magnitude alone must carry the signal, so the learned behavior survives to GRPO and inference.

reference vs. shown box



shift given in box-relative fractions